

A Two-Level Multisensor Fusion Network With Incremental Learning for Insulation Defect Diagnosis in Gas-Insulated Switchgear

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Abstract—Gas-insulated switchgear (GIS) insulation defect diagnosis faces persistent challenges due to heterogeneous multisensor signals, the lack of structured fusion across sensor types, and severe catastrophic forgetting under source-free streaming conditions. To address these issues, this article proposes a two-level fusion multisensor network with incremental learning (TFMNIL) for on-site diagnosis. Unlike conventional one-stage fusion strategies, the proposed framework separates multisensor fusion into channel-level fusion of homogeneous sensors to preserve consistent physical characteristics, and feature-level fusion of heterogeneous sensors to reduce cross-modal semantic discrepancies and redundancy. To support continual adaptation under evolving operating conditions, an incremental domain adaptation strategy is further developed. It integrates dynamic prototype updating to compensate for feature drift with classifier-weight-anchored cross-domain contrastive alignment, enabling stable class-level supervision without access to source data. Experiments on large-scale real-world multisensor GIS datasets show that TFMNIL achieves a diagnostic accuracy of 95.86% and maintains strong adaptability to newly emerging fault types in practical on-site scenarios.

Index Terms—Domain adaptation, gas-insulated switchgear (GIS), incremental learning, insulation defect diagnosis, multisensor fusion.

NOMENCLATURE

Symbols Definitions

$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$ Multichannel matrix.
 n Number of channels.

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K Total number of classes.
 H, W Height and width of the feature map x_n .
 Z_n Channel descriptor.
 σ_1, σ_2 Activation functions.
 $\mathbf{W}_1, \mathbf{W}_2$ Parameter matrices.
 $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$ Learned weights.
 $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \dots, \tilde{\mathbf{x}}_n]$ Fused sample.
 $\mathbf{X}_{avg}$ Feature vector of global average pooling.
 $\mathbf{X}_{max}$ Feature vector of max pooling.
 $\mathbf{X}$ Feature map.
 $M(\mathbf{X})$ Spatial attention map.
 $\sigma(\cdot)$ Sigmoid function.
 $\tilde{\mathbf{X}}$ Final enhanced feature map.
 $\mathbf{F}_1, \mathbf{F}_2, \text{ and } \mathbf{F}_3$ Different FC layers.
 $\mathbf{V}_1, \mathbf{V}_2, \text{ and } \mathbf{V}_3$ Weighting vectors.
 $\mathbf{F}_{final}$ Fused feature representation.
 S_{T_n} Threshold for identifying shared classes at the current time step.
 $\mathbf{F}_i^t$ Feature vectors of the i th sample during the learning of task t .
 N_{T_n} Number of the target domain samples.
 $\mathcal{L}(\bullet)$ Loss function of the model.
 $C(\bullet)$ Classifier of the model.
 $\Delta_c^{u \rightarrow t}$ Prototype shift.
 α_i Weight of sample i .
 c_k^t True prototype of class c after learning task t .
 α, β, η Balancing hyperparameters.
 $\mathcal{L}_{ce}$ Cross-entropy loss.
 $\mathcal{L}_{t, epr}$ Enhanced prototype replay loss.
 $\mathcal{L}_{fdk}$ Feature knowledge distillation loss.
 $\mathcal{L}_{cdc}$ Cross-domain contrastive loss.
 $\mathcal{L}$ Overall optimization objective.

I. INTRODUCTION

GAS-INSULATED switchgear (GIS) is a critical component in modern power systems due to its compact design, environmental compatibility, high reliability, and long maintenance intervals [1]. However, partial discharge (PD) is almost inevitable throughout the GIS lifecycle—from design and manufacturing to operation. Persistent PD can degrade insulation, leading to flashovers, breakdowns, or even catastrophic failures,

posing safety risks and economic losses [2]. Accurate diagnosis of PD caused by insulation defects are therefore essential to ensure GIS reliability [3].

In recent years, deep learning-based approaches have been widely investigated for GIS insulation defect diagnosis. Convolutional neural networks (CNNs) [4], [5], [6], [7] and long short-term memory networks [8], using ultrahigh-frequency (UHF) or acoustic emission (AE) signals, have shown strong capability in extracting spatial and temporal features of PD. However, their dependence on single-sensor data makes them vulnerable to electromagnetic interference in ON-site environments, often resulting in false alarms.

To overcome these limitations, multisensor fusion techniques combining UHF, and AE have gained attention. Notably, Dempster-Shafer evidence theory-based fusion methods have achieved encouraging results [9], [10]. In addition, chemical emission (CE) has emerged as an indispensable complementary tool, leading to exploratory studies on electrochemical joint detection [11], [12]. Yet, most efforts remain at the detection level, and research on deep multisensor fusion and deep feature learning for GIS diagnosis is still limited. Although deep fusion networks have achieved remarkable progress in other domains [13], [14], [15], [16], their direct application to GIS PD diagnosis remains challenging due to the complex and heterogeneous nature PD signals. Moreover, discrepancies between laboratory and field data hinder model generalization under multisource heterogeneous conditions.

To improve generalization under small-sample conditions, domain adaptation [17] has been widely applied in GIS PD diagnosis [18], [19], [20]. These methods have enabled partial adaptation from laboratory to field; however, several challenges remain. First, existing fusion-based diagnostic algorithms still struggle to jointly capture the electrochemical features of GIS insulation defects and to effectively integrate multisensor information for reliable diagnosis. Second, most fusion strategies operate only at data- or decision-level fusion, failing to fully exploit the complementary features among different sensors. Finally, current domain adaptation models lack incremental learning capability. As fully plastic architectures, they often suffer from catastrophic forgetting when new defect types arise, since retraining on all historical data is infeasible in streaming ON-site environments.

Incremental learning [21], [22], [23] provides a promising solution, enabling models to adapt to evolving data distributions without accessing past data. While not yet applied to GIS diagnosis, incremental methods—such as regularization-based, parameter isolation, and exemplar replay strategies [24], [25]—have shown success in other domains. In mechanical fault diagnosis, incremental frameworks allow models to recognize new fault types without full retraining [26], [27], [28], [29], [30]. Prototype-guided continual adaptation [31] further improved this by aligning target prototypes with source centroids, while multiprototype alignment strategies [32] enhanced adaptability and robustness. These advances provide valuable insights for developing GIS defect diagnosis models under dynamic, real-world conditions.

To this end, we propose a two-level fusion multisensor network with incremental learning (TFMNIL) for GIS insulation defect diagnosis under continuously evolving field conditions. Specifically, UHF, AE, and CE signals are integrated within a hierarchically structured fusion framework, in which intrasensor consistency is preserved before cross-modal interaction is

performed, reflecting the physical and statistical characteristics of GIS monitoring systems. Meanwhile, an incremental, source-free domain adaptation mechanism is introduced to address catastrophic forgetting and representation drift during long-term deployment, enabling stable diagnosis as operating conditions and defect types evolve. The main contributions are as follows.

- 1) A TFMNIL framework is proposed, which formulates multisensor fusion and incremental domain adaptation within a unified diagnostic pipeline, explicitly targeting nonstationary, source-free ON-site GIS monitoring scenarios.
- 2) A hierarchical two-level multisensor fusion strategy is developed, which explicitly decouples homogeneous-sensor fusion based on the physical characteristics of GIS sensing. Channel-level fusion preserves intrasensor physical consistency and enhances complementary channel responses, while feature-level fusion models cross-modal semantic correlations and suppresses intersensor redundancy, yielding more stable and discriminative representations.
- 3) An incremental domain adaptation mechanism with dynamic prototype correction is introduced, in which prototype drift induced by continual feature evolution is explicitly modeled and compensated, and classifier weights are reinterpreted as class-level anchors for cross-domain contrastive alignment, enabling robust adaptation and knowledge retention without accessing source data.
- 4) A large-scale, real-world multisensor GIS PD dataset is constructed from long-term substation monitoring, encompassing diverse voltage levels, manufacturers, and sensor configurations. The dataset captures realistic domain shifts induced by evolving operating conditions and, thus, provides a practical benchmark for evaluating continual and multisensor GIS PD diagnosis methods.

II. PROPOSED METHODOLOGY

This article proposes a TFMNIL for GIS insulation defect diagnosis under realistic operating conditions, where data distributions evolve over time and defect categories may expand without access to historical source data. As illustrated in Fig. 1, TFMNIL consists of three tightly coupled components: 1) a GIS multisensor data acquisition module; 2) a hierarchical two-level fusion network for structured representation learning from heterogeneous PD signals; 3) a source-free class-incremental learning module that continuously updates class knowledge under domain shift, i.e., changes in data distributions between historical source domains and evolving field conditions, while mitigating catastrophic forgetting. By jointly coordinating feature fusion and incremental adaptation within a unified pipeline, the proposed framework enables scalable and robust GIS insulation defect diagnosis under nonstationary field environments.

A. Deep Two-Level Multisensor Fusion Network

To exploit complementary information from heterogeneous GIS sensors while respecting their distinct physical sensing mechanisms, we design a physics-aware two-level multisensor

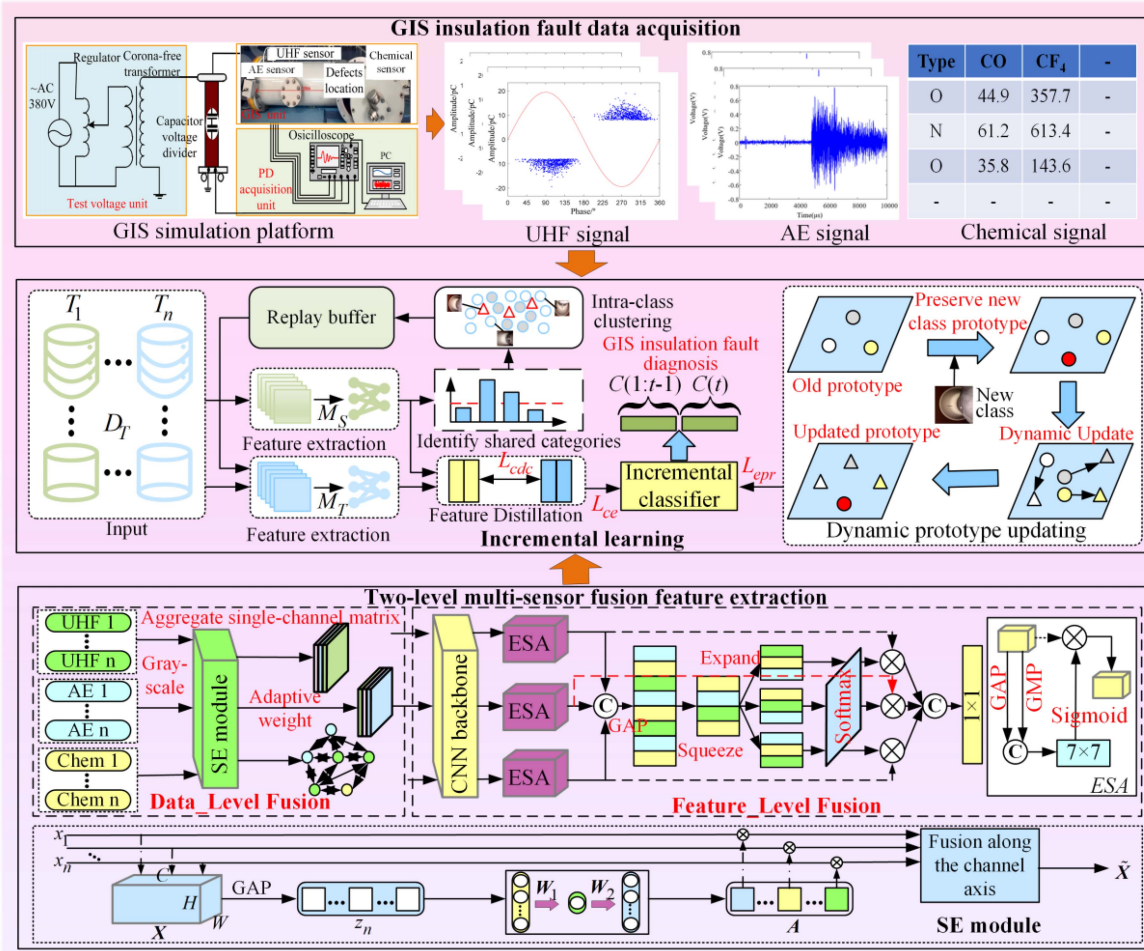


Fig. 1. Overall structure of the proposed TFMNIL.

fusion framework. Unlike single-stage fusion schemes that directly mix modalities, the proposed structure decomposes fusion into homogeneous-level aggregation and heterogeneous-level integration, following the signal formation characteristics of UHF, AE, and chemical sensors. At the first level, homogeneous sensor data are fused to preserve sensor-consistent physical characteristics and suppress redundant or noisy responses. At the second level, heterogeneous features extracted from UHF, AE, and CE are integrated through a deep feature fusion network that explicitly models cross-modal semantic discrepancies.

1) Channel-Level Fusion of Homogeneous Sensors: At the first level, data from homogeneous sensors are fused to preserve comprehensive signal characteristics while maintaining their shared physical response patterns, as shown in Fig. 1. Each sensor's raw PD signal $x_i \in \mathbb{R}^{H \times W}$ first converted into a single-channel matrix. Specifically, each single-channel matrix is obtained by directly reshaping the synchronously acquired PD signal, without any additional preprocessing. These are aggregated into a multichannel matrix $\mathbf{X} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{H \times W \times n}$. Since homogeneous sensors share similar discharge coupling paths, fusion is formulated as a channel reweighting problem rather than a semantic alignment task. Accordingly, a squeeze-and-excitation (SE) mechanism [3] is adopted to adaptively emphasize informative channels while preserving spatial structure. The channel descriptor is computed via global average

pooling:

$$z_n = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_n(i, j) \quad (1)$$

where H and W represent the height and width of the feature map x_n , respectively, and n denotes the number of channels.

Then, the descriptor is passed through a two-layer fully connected (FC) network to learn channel weights

$$\mathbf{A} = \sigma_2 [\mathbf{W}_2 \sigma_1 (\mathbf{W}_1 \mathbf{Z})] \quad (2)$$

where σ_1 and σ_2 denote the activation function, $\mathbf{Z} = [z_1, z_2, \dots, z_n] \in \mathbb{R}^{H \times W \times n}$ denotes the channel descriptor matrix, and $\mathbf{W}_1, \mathbf{W}_2$ are the parameter matrices.

Finally, the learned weights $\mathbf{A} = [a_1, a_2, \dots, a_n] \in \mathbb{R}^{H \times W \times n}$ are multiplied with the spatial information $\mathbf{X}$ to obtain the new embedded vector, i.e., the fused sample $\tilde{\mathbf{X}} = [\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n]$. The computation is as follows:

$$\tilde{x}_n = a_n \otimes x_n \quad (3)$$

where $\otimes$ denotes elementwise multiplication.

2) Feature-Level Fusion of Heterogeneous Sensors: The second-level fusion integrates features from heterogeneous sensors with fundamentally different signal generation mechanisms. Direct concatenation across modalities is avoided due to

Algorithm 1: Enhanced Spatial Attention (ESA).

Input: Feature $\mathbf{X}$, reduction ratio $r = 16$, spatial conv kernel $k_s = 7$ (stride = 1, padding = $\lfloor k_s/2 \rfloor$, dilation = 1).
Output: Enhanced feature map $\tilde{\mathbf{X}}$

- 1: # Channel Attention
- 2: Global average pooling $\mathbf{X}_{avg} \leftarrow GAP(\mathbf{X})$;
- 3: Global max pooling $\mathbf{X}_{max} \leftarrow GMP(\mathbf{X})$;
- 4: # Shared MLP with ReLU and Sigmoid activations:
- 5: $\mathbf{X}_{avg} \leftarrow \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{X}_{avg}))$ where $\delta = \text{ReLU}$, $\sigma = \text{Sigmoid}$;
- 6: $\mathbf{X}_{max} \leftarrow \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{X}_{max}))$;
- 7: $\mathbf{s} \leftarrow \sigma(\text{MLP}(\mathbf{X}_{avg}) + \text{MLP}(\mathbf{X}_{max}))$;
- 8: Channel-wise reweighting $\mathbf{X} \leftarrow \mathbf{s} \otimes \mathbf{X}$;
- 9: # Spatial Attention (7×7 conv over pooled maps)
- 10: average across channels $\mathbf{F}_{avg} \leftarrow \text{Mean}_c(\mathbf{X})$;
- 11: max across channels $\mathbf{F}_{max} \leftarrow \text{Max}_c(\mathbf{X})$;
- 12: Concatenate pooled maps:
 $\mathbf{F} \leftarrow \text{Concat}(\mathbf{F}_{avg}, \mathbf{F}_{max})$;
- 13: Spatial attention map:
 $\mathbf{M}(\mathbf{X}) \leftarrow \sigma(\text{Conv}(\mathbf{F}; \text{in} = 2, \text{out} = 1, k = k_s))$;
- 14: Spatial reweighting $\tilde{\mathbf{X}} \leftarrow \mathbf{X} \otimes \mathbf{M}(\mathbf{X})$;
- 15: **Return:** $\tilde{\mathbf{X}}$.

semantic mismatch. Instead, heterogeneous fusion is formulated as a cross-modal spatial consensus problem. Each modality is first encoded by an independent CNN backbone. Feature interaction is then regulated by an enhanced spatial attention (ESA) module [33], applied only at the heterogeneous fusion stage. Unlike convolutional block attention module, ESA is deliberately restricted to enforcing spatial consistency across modalities, while channel calibration is already addressed in the first fusion level. Specifically, ESA exploits global average pooling (GAP) and global max pooling (GMP) to capture complementary statistical descriptors of spatial activations

$$\mathbf{X}_{avg} = \text{GAP}(\mathbf{X}) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}_{i,j} \quad (4)$$

$$\mathbf{X}_{max} = \text{GMP}(\mathbf{X}) = \max \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}_{i,j}. \quad (5)$$

The resulting descriptors are processed by a shared multi-layer perceptron to generate channel attention weights. The reweighted feature map is then passed to the spatial attention branch. Specifically, average pooling and max pooling are first applied along the channel dimension, and the resulting feature maps are concatenated and convolved with a 7×7 kernel followed by a Sigmoid activation to produce the spatial attention map $\mathbf{M}(\mathbf{X})$

$$\mathbf{M}(\mathbf{X}) = \sigma(\text{Conv}(\text{Concat}(\mathbf{X}_{avg}, \mathbf{X}_{max}))) \quad (6)$$

where $\sigma(\cdot)$ is the Softmax function, Conv is the convolution operation, and Concat is the concatenation operation.

Finally, spatial reweighting is performed via elementwise multiplication between the attention and the input feature maps

$$\tilde{\mathbf{X}} = \mathbf{X} \otimes \mathbf{M}(\mathbf{X}). \quad (7)$$

In the multimodal fusion strategy, ESA-processed features from different backbones are concatenated and pooled to form a unified channel descriptor vector $\mathbf{V}$. The formulation below

illustrates three backbones but is scalable to additional heterogeneous sensor types. The computation is given as

$$\mathbf{V} = \text{GAP}(\text{Concat}(\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3)) \quad (8)$$

where GAP is global average pooling; $\mathbf{F}_i$ is the features extracted from different types of sensors at the first level.

The vector $\mathbf{V}$ is first passed through a FC layer for global feature extraction, then passed through three separate FC layers to learn channelwise importance for each feature map, yielding three weighting vectors $\mathbf{V}_1$, $\mathbf{V}_2$ and $\mathbf{V}_3$

$$\mathbf{V}_i = \text{FC}_i(\text{Ex}(\text{Sq}(\mathbf{V}))) \quad (9)$$

where FC_i represents different FC layers; Ex denotes dimensional expansion, while Sq denotes dimensional reduction. Finally, the reweighted feature maps are summed to obtain the fused feature representation $\mathbf{F}_{\text{final}}$ as follows:

$$\mathbf{F}_{\text{final}} = \text{Concat}(\mathbf{F}_1 \otimes \sigma(\mathbf{V}_1), \mathbf{F}_2 \otimes \sigma(\mathbf{V}_2), \mathbf{F}_3 \otimes \sigma(\mathbf{V}_3)). \quad (10)$$

B. Incremental Learning Under Domain Shift

We propose an incremental domain adaptation strategy for continual learning under evolving ON-site conditions, addressing catastrophic forgetting [22], label space mismatch, and domain shift without relying on source-domain data or parameter expansion. A dynamic prototype update mechanism is embedded to preserve and progressively refine class representations. This design enables the model to retain previously learned defect knowledge while continuously integrating new information under changing operational conditions. To enhance generalization to unseen defect types, classifier weights learned from the source domain are reinterpreted as class-level semantic anchors, rather than fixed decision parameters. A cross-domain contrastive alignment module then enforces domain-invariant feature alignment around these anchors, ensuring stable class boundaries even in the absence of source samples.

In source-free incremental scenarios, identifying shared classes between the current step and the source domain is crucial. Since target samples are more likely to belong to shared classes, a threshold method determines shared classes. The cumulative prediction probability of all target samples under the source model at the current step is computed as follows:

$$P_k = \sum_{i=1}^{N_{T_n}} \delta_k(H_s(G_s(x_i))) \quad (11)$$

$$P_k = \frac{P_k - \min(P)}{\max(P) - \min(P)}$$

where P_k denotes the predicted probability, N_{T_n} denotes the number of target domain samples, and $\mathbf{P}$ represents the prediction probability vector over all K classes. After computing the cumulative probabilities, a threshold is set to identify the shared classes at the current time step, which is defined as follows:

$$S_{T_n} = \{k \mid P_k > \text{thre}, \forall k \in K\}. \quad (12)$$

After identifying the shared classes, pseudolabels for the i th target domain sample can be generated by computing the nearest class centroid

$$\hat{y}_i = \underset{k}{\text{argmin}} \text{Dis}(\mathbf{F}_{\text{final}}, c_k) \quad (13)$$

where $Dis(z_i, c_k)$ denotes the distance between two vectors. The class centroid c_k is calculated as follows:

$$c_k = \frac{\sum_{i=1}^{N_{T_n}} p_i^k \mathbf{F}_{final i}^k}{\sum_{i=1}^{N_{T_n}} p_i^k}, \quad k \in S_{T_n}. \quad (14)$$

During incremental learning, prototypes of all previously learned classes are retained and replayed through the classifier to preserve prior decision knowledge. At task t , the prototype replay-based loss is defined as

$$\mathcal{L}_{t,pr} = \sum_{i=1}^{t-1} \mathcal{L}(C(c_{ki}; \delta_t), \hat{y}_i) \quad (15)$$

where $\mathcal{L}(\bullet)$ denotes the loss function of the model, and $C(\bullet)$ represents the model's classifier.

However, as the feature extractor is continuously updated, stored prototypes may gradually deviate from the true class representations in the evolving feature space, resulting in prototype drift [32]. To address this, we propose a dynamic prototype updating strategy that estimates and corrects prototype shifts using information from current tasks inspired by [34]. Let c_k^t denote the true prototype of class c after task t , where c belongs to classes learned up to task u . Due to continuous updates of the feature extractor, a discrepancy arises between the stored prototype c_k^u and the true prototype c_k^t in the current feature space. We formally define this discrepancy as the prototype drift accumulated from task u to task t

$$\Delta_c^{u \rightarrow t} = c_k^t - c_k^u. \quad (16)$$

However, during the learning of task t , samples from class c are no longer accessible, making it impossible to directly compute c_k^t or the drift term in (16). Instead of attempting exact recovery, we approximate the prototype drift by extrapolating the observed feature evolution of current-task samples, assuming that the feature extractor induces a globally smooth and bounded transformation across tasks. Specifically, for a sample i belonging to the current task class c , its feature displacement induced by learning task t is defined as

$$\partial_i^{t-1} = \mathbf{F}_i^t - \mathbf{F}_i^{t-1}, \hat{y}_i \in c \quad (17)$$

where $\mathbf{F}_i^t$ and $\mathbf{F}_i^{t-1}$ represent the feature vectors of the i th sample in class c during the learning of task $t-1$ and after task t , respectively.

We assume that the feature extractor update induces a globally smooth and bounded transformation, such that samples that are close in the feature space experience similar representation shifts. Under this assumption, the drift of an old class prototype can be approximated by aggregating the observed feature displacements of current-task samples, weighted by their proximity to the stored prototype of class c . Accordingly, the local prototype drift from task $t-1$ to task t is estimated as

$$\Delta_c^{t-1 \rightarrow t} = \frac{\sum_{(i, \hat{y}_i \in c)} \alpha_i \bullet \partial_i^{t-1 \rightarrow t}}{\sum_{(i, \hat{y}_i \in c)} \alpha_i} \quad (18)$$

$$\alpha_i = e - \frac{\|\mathbf{F}_i^{t-1} - c_k^{t-1}\|_2^2}{2\sigma^2} \quad (19)$$

Where α_i denotes the weight of sample i , and $\partial_i^{t-1 \rightarrow t}$ denotes the characteristic evolution variable of sample i from time $t-1$ to t . By sequentially applying the estimated local drift at each

incremental step, the stored prototype is progressively aligned with the evolving feature space. Accordingly, the definition of the prototype dynamic update strategy is given as follows:

$$c_k^t = \underbrace{c_k^u + \Delta_c^{u \rightarrow u+1} + \dots + \Delta_c^{t-1 \rightarrow t}}_{c_k^{u+1}}. \quad (20)$$

Importantly, this update does not aim to recover an exact prototype but to reduce accumulated misalignment over incremental steps, ensuring semantic consistency under bounded feature drift. Different from expandable subspace or ensemble-based methods [34], our approach preserves all classes within a single shared embedding space and regulates prototype evolution directly, avoiding task isolation and uncontrolled parameter growth. Finally, the loss function based on the enhanced prototype replay is defined as follows:

$$\mathcal{L}_{t, epr} = \sum_{i=1}^{t-1} \mathcal{L}(C(c_{ki}^t; \delta_t), \hat{y}_i). \quad (21)$$

Sample replay preserves prior knowledge by capturing intraclass variations and interclass relationships, but domain shifts can introduce noise into pseudo- and soft labels. To address this, we propose a cross-domain contrastive alignment strategy, enhancing inference accuracy. By regularizing the feature space during prototype alignment, it preserves prior knowledge and mitigates catastrophic forgetting as new classes are introduced. In class-incremental source-free domain adaptation, classifier weight vectors from a pretrained model serve as approximate class centroids. These weights C_s as prototypes to construct class-level contrastive pairs, allowing computation of a contrastive loss that maintains class separability despite the absence of source samples

$$C_s \rightarrow [w_1, w_2, \dots, w_k] \quad (22)$$

where w_k denotes the weight vector corresponding to the k th class in the classification layer. Accordingly, the cross-domain contrastive loss can be formulated as

$$\mathcal{L}_{cdc} = -\frac{1}{|\mathcal{R}_{T_n}|} \sum_{i=1}^{|\mathcal{R}_{T_n}|} \log \frac{\exp(G_t(r_i) w_{\hat{y}_i} / \tau)}{\sum_{j=1}^K \exp(G_t(r_i) w_j / \tau)} \quad (23)$$

where τ denotes the temperature coefficient, and $|\mathcal{R}_{T_n}|$ represents the number of prototypes stored in the replay buffer.

C. Training Process

Firstly, when learning the new task t , cross-entropy loss [3] is adopted as the classification loss, which is defined as follows:

$$\mathcal{L}_{ce} = -\frac{1}{N_{T_n}} \sum_{i=1}^{N_{T_n}} \hat{y}_i \log \delta(H_t(G_t(x_i))). \quad (24)$$

To reduce feature-level forgetting, a feature knowledge distillation constraint is applied to the extractor, preserving learned feature stability and mitigating prototype shifts. An L_2 norm enforces consistency between current and previous extractors, and the feature knowledge distillation loss for task t is defined as

$$\mathcal{L}_{fkd} = \|\mathbf{F}_t(\tilde{x}_t; \theta_t) - \mathbf{F}_{t-1}(\tilde{x}_t; \theta_t)\|_2^2. \quad (25)$$

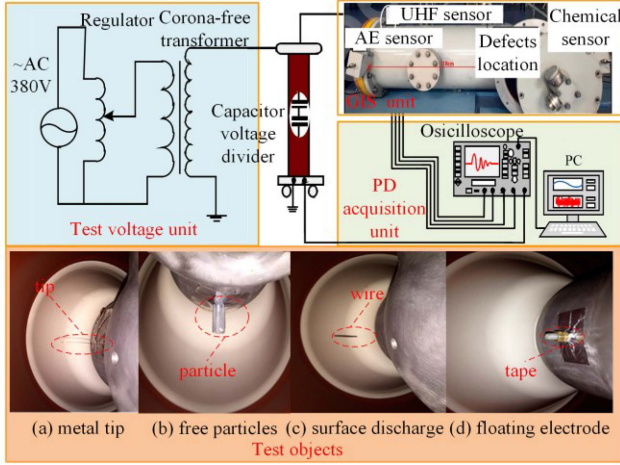


Fig. 2. 252 kV GIS simulation platform.

Algorithm 2: The Proposed TFMNIL Algorithm.

Input: Samples at the current time step $D_t = \{X_i\}_{i=1}^{N_{T_n}}$, replay buffer $\mathcal{R}_{T_{n-1}}$, pre-trained source model M_s .
Output: Target domain model M_t

- 1: Identify shared classes according to (12);
- 2: **for** $epoch = 1, 2, 3, \dots, E$ **do**
- 3: Compute cross-entropy loss according to (24);
- 4: Compute replay loss according to (21);
- 5: Compute feature knowledge distillation loss according to (25);
- 6: Compute contrastive alignment loss according to (23);
- 7: Optimize the target model according to (26);
- 8: **end for**
- 9: **Return:** Return the target model M_t at the current time step.

By integrating all the aforementioned loss components, the overall optimization objective of this study can be formulated as follows:

$$\min_{\{\theta_{G_t}, \theta_{H_t}\}} \mathcal{L} = \min_{\{\theta_{G_t}, \theta_{H_t}\}} \mathcal{L}_{ce}(\theta_{G_t}, \theta_{H_t}, \mathbf{x}) + \mu \mathcal{L}_{t, epr}(\theta_{G_t}, \theta_{H_t}, \mathbf{r}) + \beta \mathcal{L}_{fkd}(\theta_{G_t}, \theta_{H_t}, \mathbf{x}) + \eta \mathcal{L}_{cdc}(\theta_{G_t}, \mathbf{r}, \mathbf{w}) \quad (26)$$

where α , β , and η are the balancing hyperparameters. The detailed algorithm flow is shown in Algorithm 2.

III. GIS INSULATION DEFECT DIAGNOSIS DATA COLLECTION

A 252 kV GIS simulation platform, as shown in Fig. 2 was built to acquire representative insulation defect samples. Internal discharge was monitored using UHF and AE sensors, complemented by chemical detection. The high-voltage circuit includes a 150 kV AC supply, high-voltage bushing, and 220 kV GIS chamber. UHF signals are recorded with an external sensor (300–2000 MHz, 20 dB) on a four-channel WavePro 254HD oscilloscope (2.5 GHz, 20 GS/s). AE signals are captured by ultrasonic sensors (20–200 kHz, 100 dB), while a micro gas sensor array monitors nine SF₆ decomposition products [11].

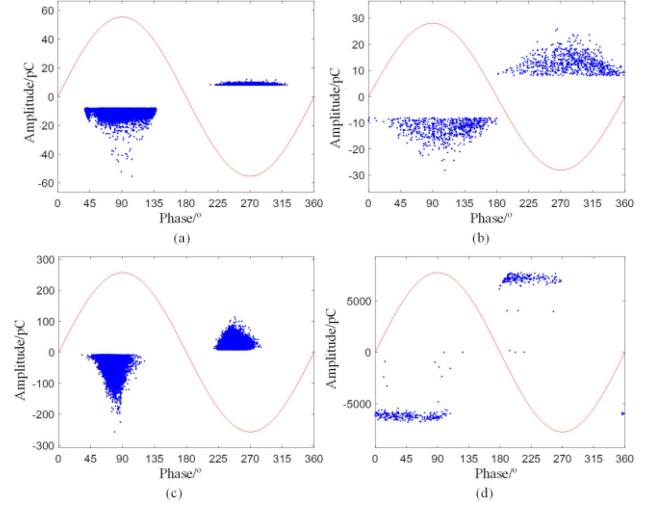


Fig. 3. UHF signals for each defect category.

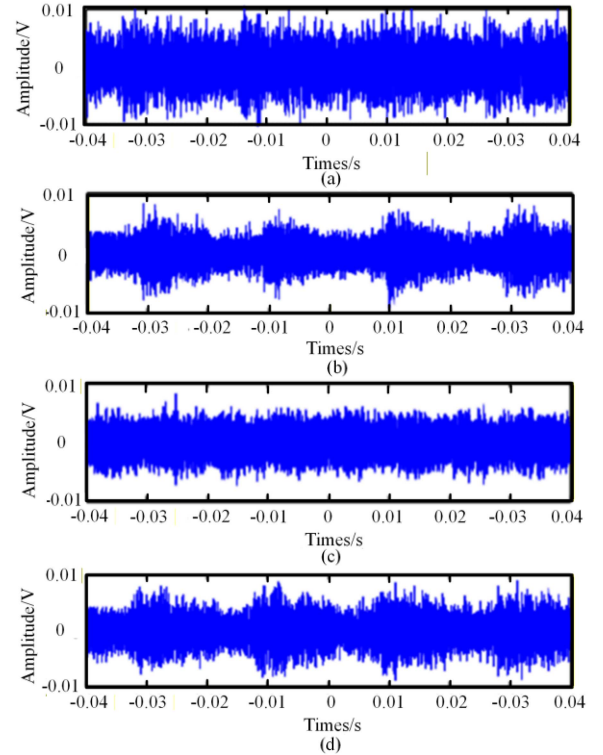


Fig. 4. AE signals for each defect category.

Four typical insulation defects—enclosure protrusion (M-type), metallic particle on the conductor (N-type), metallic particle on the insulator (O-type), and enclosure floating potential (Q-type)—were fabricated for PD experiments, each repeated three times. Voltage was gradually increased to determine inception voltage U_0 , held for 5 min, then raised to 1.2, 1.5, and 2 U_0 with specified durations. UHF and AE sensors recorded PD signals, with background noise below 0.7 pC. Tests were performed at multiple defect locations to enhance diversity, yielding 500 sample sets per defect type. Representative UHF and AE signals are shown in Figs. 3 and 4.

TABLE I
OVERALL COMPOSITION OF THE CONSTRUCTED DATASET

Domain	Defect type					Total
	Metal tip	Free particles	Surface discharge	Floating electrode	Air gap	
Laboratory experiments	500	500	500	500	—	2000
On-site measurements	114	120	118	106	104	562

As shown in Figs. 3 and 4, the metal tip defect exhibits concentrated, high-amplitude UHF pulses at specific phase angles and strong, regular AE signals. Free particles show scattered UHF pulses over a wide phase range and irregular, moderate AE responses. Surface discharge presents phase-periodic UHF pulses with moderately strong, regular AE signals, while the floating electrode defect exhibits weak and irregular patterns in both modalities. The comparison indicates that HUF patterns reflect the electrical characteristics of each defect, whereas AE signals reveal their acoustic responses.

Field data were collected from long-term operational records and ON-site monitoring of 110–750 kV SF₆ GIS units deployed across multiple substations, focusing on equipment exhibiting detectable PD activity. Priority monitoring captured the full PD lifecycle—from initial occurrence to system shutdown and disassembly—allowing internal defects to be confirmed and reliably labeled. This produced 562 samples combining SF₆ decomposition gas data with UHF and AE measurements, supplementing laboratory-induced defects with insulator air gap faults (P-type).

Data were accumulated over multiple years, across GIS units of different vendors and voltage levels. All features were Z-score normalized to mitigate device-specific discrepancies. Owing to its diversity in equipment types, sensor configurations, and operating conditions, the resulting dataset provides a representative and challenging testbed for evaluating the generalization capability of GIS PD diagnostic models, as summarized in Table I. The dataset was, therefore, designed to support scenario-driven incremental evaluation rather than fixed benchmark protocols. Notably, heterogeneous sensor data are organized at the level of inspection instances rather than continuous streams. Transient UHF/AE signals characterize recent discharge activity, while SF₆ indicators reflect accumulated insulation conditions over a preceding period. Each modality is first mapped to a temporally abstracted state-level representation, and feature-level fusion is performed only after this abstraction, thereby avoiding incompatible time-scale mixing and the resulting bias.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

Experiments were conducted on a system with an Intel Core i9-10900K CPU, NVIDIA RTX 3090 GPU (24 GB), and 64 GB RAM, using Python 3.8, PyTorch 1.2.0, and CUDA 10.0. Models were trained with SGD (learning rate 0.001, batch size 64), and all results are averaged over three runs. For comparison with replay-based methods, a fixed memory of 20 samples per old class was used. Rather than jointly tuning all loss weights, μ and β are fixed to 1 across all datasets, while only a single parameter γ is adjusted. Crucially, the selection of γ is performed using source-domain validation data only, without any access

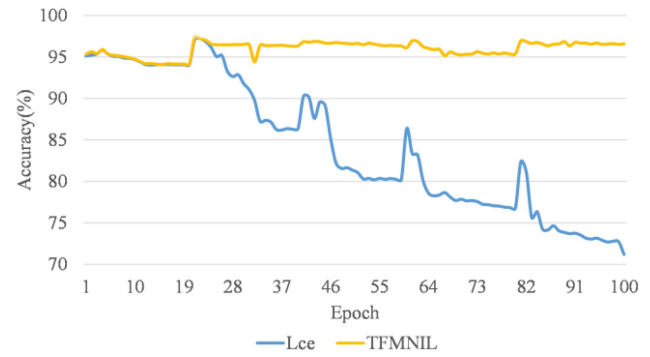


Fig. 5. Comparison of diagnosis accuracy trends.

TABLE II
DIAGNOSTIC ACCURACY USING DIFFERENT SENSOR COMBINATIONS

Sensor Configuration	Accuracy (%)	Statistical Result
UHF	94.19 ± 0.47	—
AE	93.77 ± 0.45	—
CE	94.06 ± 0.41	—
UHF+AE	94.82 ± 0.36	$p < 0.05$ versus UHF, AE
UHF+CE	94.93 ± 0.34	$p < 0.05$ versus UHF, CE
AE+CE	94.68 ± 0.38	$p < 0.05$ versus AE, CE
UHF+AE+CE	95.86 ± 0.29	$p < 0.05$ versus all dual-sensor cases

to target-domain performance. A grid search yields $\mu = 1$, $\beta = 1$, and $\gamma = 0.1$. For all experiments, a ResNet18 backbone was adopted as the base feature extractor. Source-domain data were split into training and reference test sets with an 8:2 ratio. Target-domain data were divided with a 4:6 ratio, where 40% of the target samples were used exclusively for incremental domain adaptation, and the remaining 60% were strictly held out as an unseen test set. Importantly, this held-out target subset was never accessed during adaptation, hyperparameter selection, or robustness analysis. To ensure conciseness of the manuscript while preserving the completeness of methodological details and experimental results, additional materials—including hyperparameter sensitivity analyses, ablation studies, detailed per-class diagnostic metrics, and a representative field case—are provided in the Electronic Companion, available at <https://zenodo.org/records/19463537> (DOI: 10.5281/zenodo.19463537) [35].

Fig. 5 illustrates the evolution of diagnosis accuracy across incremental steps. When trained using only the cross-entropy loss, the model exhibits severe catastrophic forgetting: although accuracy initially improves with the introduction of new classes, performance rapidly degrades as previously learned knowledge is overwritten. In contrast, TFMNIL maintains smooth and stable performance trajectories across all incremental stages, despite the joint optimization of multiple loss components.

B. Diagnostic Performance of Multisensor Fusion

The proposed TFMNIL was evaluated under different sensor combinations, and the diagnostic performance is summarized in Table II. As shown in Table II, single-sensor models yield overall accuracies ranging from 93.77% to 94.19%. Dual-sensor fusion consistently improves performance, achieving accuracies between 94.68% and 94.93%. The joint fusion of UHF, AE, and CE achieves the highest accuracy (95.86% ± 0.29). In addition, paired statistical tests were conducted between sensor configurations under identical experimental settings. The results



Fig. 6. Confusion matrices for single- and multisensor methods. (a) UHF. (b) AE. (c) CE. (d) UHF+AE+CE.

TABLE III
EFFECT OF CLASS-BALANCING STRATEGIES

Method	Overall Accuracy(%)	P-type F1 (%)	BWT (%)
Baseline (CE)	95.86	93.32	95.14
+ Focal Loss	96.02	94.51	94.97
+ Prototype Reweighting	96.11	95.03	95.21

indicate that the performance gain of the proposed three-sensor fusion over each dual-sensor baseline is statistically significant at the $p < 0.05$ level.

Fig. 6 shows confusion matrices for single- and multisensor methods. Single-sensor models display variable performance; for instance, UHF poorly detects insulator surface discharges, while AE and CE alone fail to fully capture all defect characteristics. In contrast, the proposed multisensor fusion consistently improves accuracy across all defect types, particularly for those poorly recognized by individual sensors. By jointly leveraging UHF, AE, and CE, the TFMNIL exploits complementary information, providing a robust and reliable framework for complex GIS insulation defect diagnosis.

To examine the influence of class imbalance, we evaluated two balancing strategies: 1) focal loss and 2) prototype reweighting based on inverse class frequency during prototype updating. The results are summarized in Table III. Both strategies slightly improve minority-class performance. Prototype reweighting yields the most balanced improvement, increasing the P-type F1-score by 1.71% without degrading BWT, indicating improved stability-plasticity balance.

To further assess deployment feasibility, we analyze the relationship between per-class sample size and F1-score stability. Empirically, classes with ≥ 30 samples exhibit variance below $\pm 1.2\%$ across three runs, whereas classes with fewer than 25 samples show noticeably higher fluctuation ($\pm 2.8\%$). These

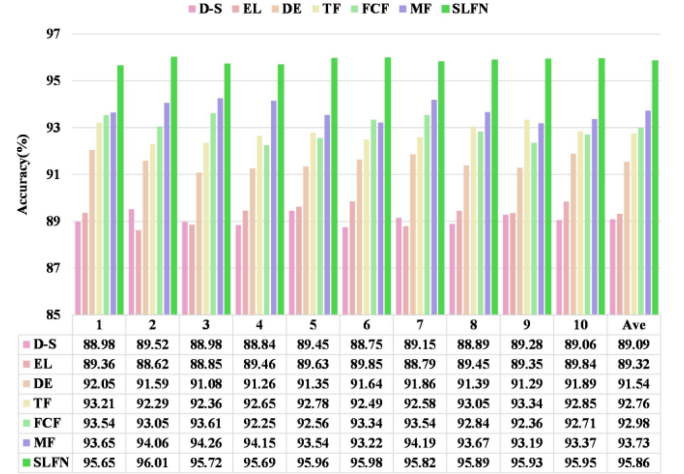


Fig. 7. Diagnostic results of GIS insulation defects using different methods.

observations suggest that a minimum of approximately 30 labeled samples per defect type is sufficient to ensure reliable field deployment under the current model capacity and feature dimensionality.

C. Comparison With Different Sensor Fusion Strategies

To validate TFMNIL, we compared it with established fusion strategies, including D-S and ensemble learning (EL), as well as four ablation variants. 1) Data-level fusion (DF), which applies only first-level homogeneous sensor fusion via SE-based channel reweighting, without heterogeneous interaction. 2) Feature-level fusion (TF), which performs only second-level heterogeneous feature fusion without prior homogeneous consolidation. 3) Feature concatenation fusion (FCF), which replaces adaptive fusion with direct concatenation. 4) Multimodal fusion without ESA (MF), which removes the spatial attention module from the deep fusion stage. The MF setting isolates the contribution of ESA in the presence of SE-based channel attention, clarifying potential redundancy between channel and spatial attention mechanisms.

As shown in Fig. 7, TFMNIL achieved 95.86% accuracy, outperforming D-S, EL, DF, TF, FCF, and MF by 6.77%, 6.54%, 4.32%, 3.1%, 2.88%, and 2.13%, respectively. The consistent performance gap between TFMNIL and both DF and TF demonstrates that neither homogeneous fusion nor heterogeneous fusion alone is sufficient; instead, their structured combination is critical for fully exploiting multisensor complementarity. Moreover, the performance gain over MF indicates that ESA provides additional discriminative capability beyond channel reweighting alone, confirming that ESA is not functionally redundant with SE but instead complements it by enhancing cross-modal spatial consistency during heterogeneous fusion.

Fig. 8 compares feature distributions across fusion strategies. TFMNIL yields more compact intraclass clusters with fewer misclassifications, reflecting stronger discriminative learning. DF maintains intrasensor consistency but lacks cross-modal interaction, leading to dispersed features, whereas TF enables heterogeneous interaction but produces unstable representations without homogeneous consolidation. By enforcing low-level physical consistency followed by high-level semantic

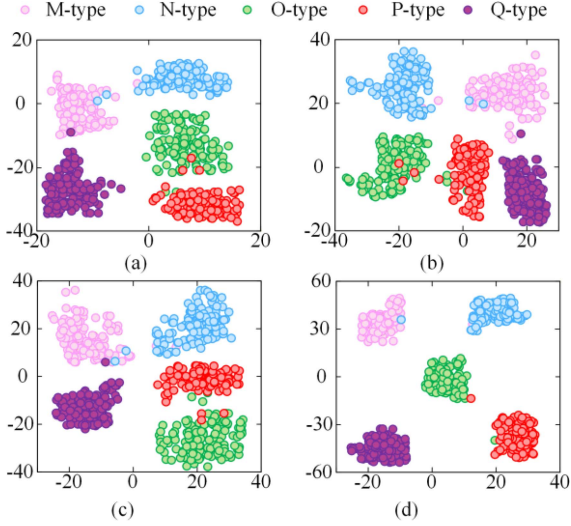


Fig. 8. Feature visualization results. (a) DF. (b) TF. (c) FCF. (d) TFMNIL.

TABLE IV
QUANTITATIVE CLUSTERING QUALITY UNDER DIFFERENT FUSION STRATEGIES

Method	IC	IS
DF	1.84	3.21
TF	1.57	3.48
FCF	1.49	3.64
TFMNIL	1.17	4.15

TABLE V
DIAGNOSTIC ACCURACY UNDER DIFFERENT DOMAIN ADAPTATION SCENARIOS

Methods	Diagnosis accuracy (%)		
	Scenario 1	Scenario 2	Scenario 3
DANN	76.82	75.68	74.54
MSDA	85.62	79.65	72.19
PADA	84.68	63.54	60.58
ALDAN	87.56	73.42	65.87
CEL	69.87	69.15	68.79
PR	75.36	73.92	70.85
CECA	86.35	84.39	83.25
TFMNIL	95.92	95.83	95.86

interaction, the proposed two-level fusion achieves more separable and robust features. This is corroborated in Table IV, where TFMNIL attains the lowest intraclass compactness (IC) and highest interclass separability (IS), reducing IC by 34.2% relative to DF and increasing IS by 29.8% over TF.

D. Comparison With Domain Adaptation Methods

To evaluate domain adaptation performance, TFMNIL was compared with DANN [18], MSDA [4], PADA, and ALDAN [36]. Three additional ablation variants were tested: 1) Cross-entropy loss (CEL): model training based solely on cross-entropy loss; 2) Prototype replay loss (PR): training with prototype replay loss only; 3) Cross-domain contrastive alignment loss (CECA): training with cross-domain contrastive alignment loss only. Table V summarizes GIS insulation defect diagnosis results across three scenarios: 1) fixed defect categories with increasing samples, 2) increased samples with stronger domain

TABLE VI
BACKWARD TRANSFER ACCURACY ACROSS INCREMENTAL LEARNING STAGES

Methods	BWT (%)				
	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
DANN	91.24	86.73	82.15	78.46	74.89
PADA	92.18	87.62	83.07	80.34	76.91
ALDAN	93.05	89.14	85.36	82.08	79.44
CEL	94.32	91.26	88.95	86.41	84.73
PR	95.01	92.87	90.54	88.72	87.09
CECA	95.68	94.12	93.07	92.35	91.84
TFMNIL	96.02	95.61	95.14	95.78	94.39

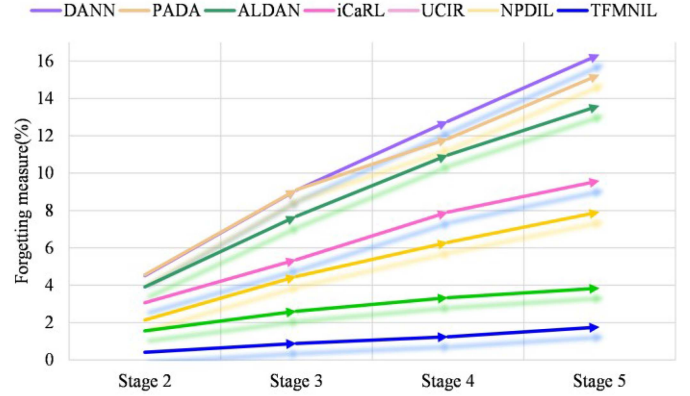


Fig. 9. Forgetting measure across incremental learning stages.

shift, and 3) increasing samples with the introduction of a new defect category. In Scenario (3), the new defect class was introduced at Step 3, after convergence on previously observed classes.

As shown in Table V, conventional unsupervised and source-free domain adaptation methods exhibit significant accuracy degradation under Scenario 3, particularly when both sample volume and category space expand. This behavior is consistent with catastrophic forgetting, as these methods tend to overadapt to newly observed target distributions at the expense of previously learned defect classes. By contrast, TFMNIL shows a marginal accuracy variation when a new defect class is introduced. Rather than indicating the absence of forgetting, this limited drop reflects a controlled incremental setting with high interclass distinctiveness and constrained parameter updates, which together mitigate abrupt representation drift.

To further assess catastrophic forgetting, we report BWT and forgetting curves across incremental stages. BWT measures performance on previously learned defect classes after each update, directly reflecting knowledge retention. As shown in Table VI, TFMNIL maintains BWT above 94% at all stages, consistent with its conservative prototype preservation and update strategy, which mitigates interference from newly introduced classes. In contrast, methods without prototype correction exhibit progressively declining BWT, indicating cumulative representation drift. Fig. 9 shows the corresponding forgetting curves, where forgetting is defined as the maximum accuracy drop on old classes relative to their peak performance. TFMNIL yields consistently low—though non-zero—forgetting, whereas competing methods display steadily increasing degradation as new defect categories are added. These results suggest that the proposed prototype correction mechanism effectively alleviates,

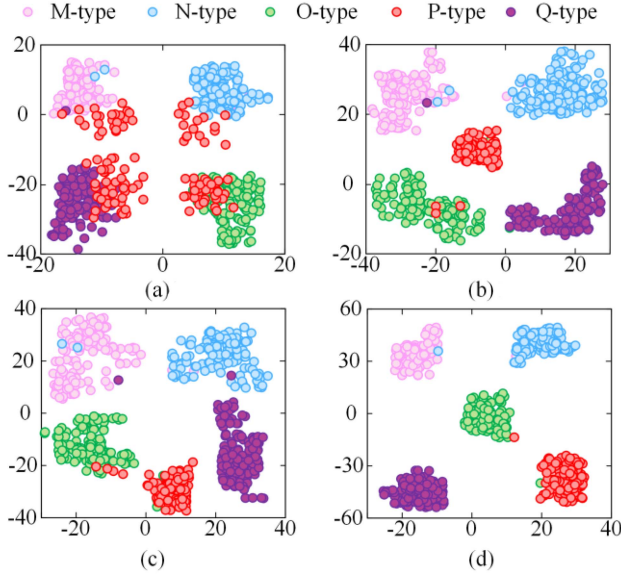


Fig. 10. Two-dimensional feature visualizations of representative methods. (a) DANN. (b) PADA. (c) ALDAN. (d) TFMNIL.

TABLE VII

ACCURACY AND STABILITY OF INCREMENTAL LEARNING METHODS

Methods	Diagnosis results		
	Accuracy (%)	Standard deviation	<i>t</i> -test (vs. TFMNIL)
iCaRL	80.56	1.23	$t = 43.08, p < 0.001$
EEIL	86.39	1.32	$t = 30.05, p < 0.001$
UCIR	88.93	0.85	$t = 42.28, p < 0.001$
NICE	90.17	0.78	$t = 28.91, p < 0.001$
EvoPrompt	92.02	0.69	$t = 24.36, p < 0.001$
EVCL	92.48	0.63	$t = 22.17, p < 0.001$
NPDIL	92.35	0.49	$t = 21.64, p < 0.001$
TFMNIL	95.86	0.41	—

though does not entirely eliminate, catastrophic forgetting in inspection-level class-incremental domain adaptation.

Fig. 10 visualizes feature distributions after incremental learning. DANN fails to separate the newly introduced air-gap discharge class, leading to severe class confusion. PADA and ALDAN improve clustering for the new class but degrade the compactness of previously learned classes, indicating knowledge forgetting. In contrast, TFMNIL maintains compact and well-separated clusters for both old and new classes.

E. Comparison With Different Incremental Learning

To assess adaptability in evolving environments, TFMNIL was compared with iCaRL, EEIL, and UCIR [28], [29]. An additional ablation variant, nonprototype dynamic incremental learning (NPDIL), was introduced by removing the dynamic prototype update to isolate its contribution. To address recent advances in incremental learning (IL), we further extended the comparison to representative state-of-the-art (SOTA) methods, including neurogenesis inspired contextual encoding (NICE), evolving parameterized prompt memory (EvoPrompt), and elastic variational continual learning (EVCL) [21], [22], [23]. The diagnostic results of different methods are shown in Table VII. As shown in Table VII, TFMNIL achieves the highest accuracy with the lowest variance, demonstrating superior stability across incremental stages. Compared with NPDIL, dynamic

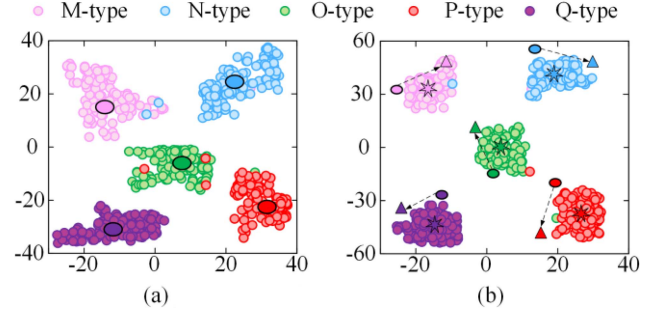


Fig. 11. Prototype distributions before and after updates. (a) Prototype before updates. (b) Prototype after updates.

TABLE VIII

COMPLEXITY AND EFFICIENCY COMPARISON OF INCREMENTAL LEARNING METHODS

Method	Complexity	FLOPs ($\times 10^9$)	Parameters ($\times 10^6$)	Training convergence time (h)	Inference latency (ms/sample)
iCaRL	$O(n) + O(m)$	42.8	17.41	0.63	0.64
EEIL	$O(n) + O(m)$	42.8	17.41	0.84	0.51
UCIR	$O(n) + O(m)$	42.8	17.41	0.75	0.53
NICE	$O(n) + O(m)$	43.2	17.85	0.89	0.56
EvoPrompt	$O(n) + O(m)$	43.7	18.31	0.93	0.59
EVCL	$O(n) + O(m)$	43.5	18.02	0.92	0.58
NPDIL	$O(n) + O(p)$	42.8	17.41	0.91	0.54
TFMNIL	$O(n) + O(p)$	42.8	17.41	0.98	0.55

prototype updating improves accuracy by 3.51%, validating its effectiveness in mitigating prototype drift and preserving class boundaries. Although recent SOTA methods such as DER++ and Adaptive-Prompt IL outperform earlier exemplar-based approaches, they exhibit greater variance under strong domain shifts and heterogeneous sensor fusion. In contrast, TFMNIL maintains both higher accuracy and stability, underscoring the advantage of jointly modeling dynamic prototypes and cross-domain contrastive alignment for non-stationary GIS PD signals. All gains are statistically significant ($p < 0.001$).

Fig. 11 further illustrates prototype evolution during incremental learning. TFMNIL produces more compact intraclass clusters and larger interclass separations than NPDIL, indicating improved discriminability. It also reduces outliers and sharpens decision boundaries, effectively preserving class structure during incremental learning. These results demonstrate that the dynamic prototype update mechanism enhances classification accuracy and supports reliable GIS insulation defect diagnosis.

Table VIII summarizes computational efficiency. Replay-based methods (e.g., iCaRL, EEIL, UCIR) and recent replay-free or prompt-based approaches incur a complexity of $O(n) + O(m)$, where n is the number of incoming samples and m the replayed exemplars. Prototype-based methods (NPDIL and TFMNIL) require $O(n) + O(p)$, with p denoting stored prototypes. TFMNIL further introduces dynamic prototype updating with cost $O(p)$, yielding $O(n) + 2O(p)$, which remains effectively $O(n) + O(p)$. As p is typically far smaller than m , TFMNIL achieves higher accuracy with lower computational overhead. Recent SOTA methods such as NICE, EvoPrompt, and EVCL add context encoders, prompt memories, or variational regularization, slightly increasing FLOPs or parameters, though overall complexity remains largely backbone-dependent and comparable. Although

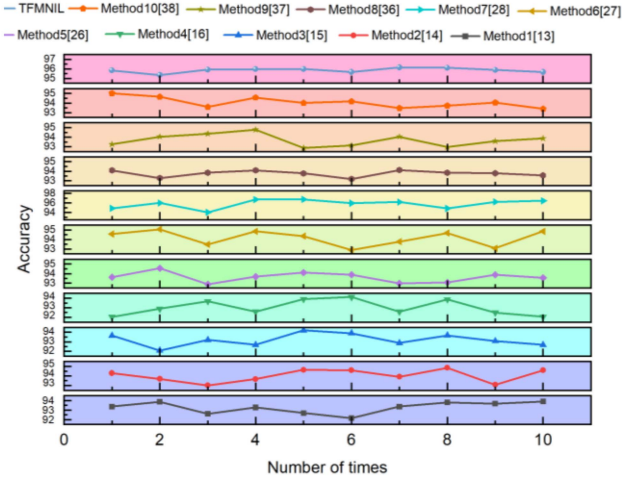


Fig. 12. Comparison results with state-of-the-art models.

TFMNIL has the same FLOPs and parameter count as other backbone-based methods, its training time is $\sim 30\%$ longer than UCIR (0.98 h versus 0.75 h). This overhead stems mainly from dynamic prototype updates and the slower convergence of the cross-domain contrastive loss, rather than increased model depth or inference complexity. The cost occurs only during offline or periodic updates; inference remains ~ 0.5 ms per sample, meeting real-time GIS monitoring requirements. In practice, the additional ~ 14 min of offline training is acceptable given intermittent model updates. In exchange, TFMNIL achieves a 6.93% accuracy gain over UCIR and markedly improved stability, prioritizing reliability and long-term performance over marginal training speed.

F. Discussion

The proposed two-level fusion network addresses key challenges in source-free incremental domain adaptation, notably label space inconsistency and catastrophic forgetting, while enabling continuous knowledge accumulation without degrading prior representations. It accommodates new insulation defect types and evolving PD characteristics without full retraining, supporting robust real-world deployment. This design is well suited to GIS PD monitoring, where signals are nonstationary and influenced by changing operating conditions and sensor distributions. Per-class evaluation shows that gains are not driven by majority classes. Even the minority P-type defect achieves an F1-score above 93%, with further improvement possible via lightweight prototype reweighting. Stable performance with as few as 30 labeled samples per class per stage indicates robustness under moderate imbalance. Although larger datasets enhance representation quality, the results suggest that extensive per-class data are not required for deployment-level accuracy.

Although source-free incremental domain adaptation has not previously been explored for GIS insulation defect diagnosis, related approaches have shown success in other industrial settings. We, therefore, compared our framework with recent SOTA methods (see Fig. 12), including multisensor fusion models (Methods 1–4) and source-free incremental domain adaptation methods (Methods 5–10), using their original configurations for fairness. TFMNIL achieves the highest average diagnostic

accuracy and the lowest variance over ten runs, demonstrating strong robustness. While many recent incremental learning methods perform well on benchmark vision datasets, their stability under heterogeneous multisensor fusion and strong domain shifts is limited. The consistent advantage of TFMNIL over both classical and recent baselines indicates that explicitly modeling sensor structure and prototype dynamics is essential for industrial PD diagnosis, beyond generic continual learning strategies. A framework robust to complex, nonstationary GIS PD data is, thus, well positioned to generalize to other industrial scenarios with less intricate fault dynamics.

Although TFMNIL preserves class knowledge through prototype replay, mitigates domain shift via contrastive alignment, and dynamically updates prototypes, its robustness may degrade under label noise and rare defects. Erroneous pseudolabels can induce incorrect updates, and underrepresented classes may weaken decision boundaries. Future work will investigate uncertainty-aware prototype selection and adaptive pseudolabeling. We also plan to extend validation to PD-based diagnostics for transformers and cables in collaboration with external laboratories, aiming to develop a unified and generalizable framework for diverse power equipment.

V. CONCLUSION

This article presented TFMNIL for diagnosing GIS insulation defects in continuously evolving on-site data scenarios. The TFMNIL addressed two core challenges in source-free domain adaptation: label space inconsistency and catastrophic forgetting. A novel two-level multisensor fusion architecture integrates UHF, AE, and CE data and features, enhancing representations and achieving 95.86% diagnostic accuracy, significantly outperforming single-sensor baselines. The adaptive fusion module captures complementary information and long-range dependencies, enabling hierarchical integration from local to global features and improving robustness.

To handle evolving sample distributions, an incremental domain adaptation strategy was employed. Prototype replay preserves class-specific knowledge, while cross-domain contrastive alignment mitigates domain shift and dynamic prototype updates correct outdated representations. These mechanisms yield tighter feature clusters, clearer decision boundaries, and improved classification under changing conditions. Future work will focus on improving robustness to rare defect types and label noise, extending the framework to PD-based diagnosis of other power equipment, and constructing a standardized, publicly accessible multisensor PD dataset based on full-scale laboratory platforms to facilitate benchmarking and broader community engagement.

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